# Lab 08

## Final Review

CSE110: Programming Language I

Practice Problems **(No Submission)**

# Strings

## Task 1

Trace the following code and write the outputs.

```
public class Trace01{
    public static void main(String[] args) {
        String course = "";
        int i = 2, j = 0, k = 18;
        course = "-->cse";
        while (i< 5) {
            k--;
            j = k;
                while (j > 12 ) {
                if (j % 2 != 0) {
                        course += "<--";
                        course =  course + i + (j / 2);
                } else {
                    course += "-->";
                        course =  course + (i % 2) + j;
                }
                System.out.println(course);
                  if (j == 14) {
                    course = "-->cse";
                }
                --j;
            }
            i++;
        }
    }
}
```

## Task 2

Write a method **modifyStrings()** that takes in three given strings **S, S1, and S2** consisting of different numbers of characters respectively, the task is to modify the string **S** by **replacing** all the **substrings S1** with the **string S2** in the string **S** and printing the modified string S.

| Sample Input | Sample Output | Explanation |
|---|---|---|
| S = "abababa"<br>S1 = "aba"<br>S2 = "a"<br><br>modifyStrings(S, S1, S2); | aba | Changing the substrings **S[0, 2]**(Referring to characters from the 0th index of **S** till the 2nd index of S and **S[4, 6]** (= S1) to the string S2 (= "a") modifies the string S to "aba". Therefore, print "aba". |
| S = "baddadda"<br>S1 = "dd"<br>S2 = "n"<br><br>modifyStrings(S, S1, S2); | banana | Changing the substrings **S[2,3]**(Referring to characters from the 2nd and 3rd index of **S**) and **S[5, 6] (= S1)** to the string **S2 (= "n")** modifies the string S to "banana". Therefore, print "banana". |

# Strings + Arrays

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## Task 3

Given an array of email addresses, print the number of valid email addresses satisfying the following conditions.

a) Each email contains an '@' character
b) There is at least one character before and after '@' character and it has to start with letter
c) There is a '.' character after the character(s) after '@' character
d) There is at least one character after '.' character

| Sample Input | Sample Output |
|---|---|
| email_list = {"abc@gmail.com", "!@cv.bd", "123cse@bracu.ac.bd"} | 1 |
| email_list = {"cse110@gmail.com", "government@cv.", "eee@bracu.ac.bd"} | 2 |

## Task 4

Write a method called isHappyNumber which takes an integer in its parameter to check whether a number is a happy number or not. If the number is a happy number then the method returns boolean true otherwise it returns boolean false. In number theory, a happy number is a number which eventually reaches 1 when replaced by the sum of the square of each digit. For instance, 13 is a happy number because $1^2 + 3^2 = 10$ and $1^2 + 0^2 = 1$. On the other hand, 4 is not a happy number because the process continues in an infinite cycle without ever reaching 1. Unhappy number ends in a cycle of repeating numbers which contains 4 .

| Sample Input | Sample Output |
|---|---|
| boolean check = isHappyNumber(82)<br><br>System.out.println(check) | true |
| boolean check = isHappyNumber(4)<br><br>System.out.println(check) | false |

## Task 5

Write a method called toDecimal which takes a binary number as a string in its parameter to convert the binary number to its decimal number and return the decimal value. After returning the decimal value, write another method called toHex which takes the converted decimal value in its parameter and calculates the hexadecimal value and then return the hex value.

| Sample Input | Sample Output |
|---|---|
| int decimal = toDecimal("1010")<br>String hex = toHex(decimal)<br>System.out.println(hex) | "A" |

## Task 6

Trace the following code and write the outputs.

```
class Trace02 {
    public static void main(String args[]) {
        int[] arr1 = {3, 1, 4, 1, 5, 9, 2};
        int[] arr2 = {10, 20, 30, 40, 50, 60, 70};
        int x = 0, y = 0;
        while (x < arr1.length - 1) {
            arr2[x] = arr1[y] * (x + 1) - arr2[y];
            y = 1;
            while (y <= x) {
                arr2[x] = arr2[x] + arr1[y] - y;
                y = y + 1;
            }
            System.out.println(arr2[x]);
            x = x + 1;
        }
        System.out.println(arr2[arr1.length - 1]);
    }
}
```

## Task 7

You are given an integer array. You need to identify all the **prime numbers** and **perfect numbers** within the array and print the **indices** along with these **numbers** from the original array.

| Sample Input | Sample Output |
|---|---|
| Sample Input:<br>int arr[] = {6, 13, 28, 17, 3, 9, 11, 23, 10, 29, 12, 7} | Prime Numbers:<br>1: 13<br>3: 17<br>4: 3<br>6: 11 |

| | |
|---|---|
| | 7: 23<br>9: 29<br>11: 7<br><br>Perfect Numbers:<br>0:  6<br>2: 28 |

## Task 8

Trace the following code and write the outputs.

```
public class tracing1 {
        public static void main(String[] args){
        int i = 1;
        int [] a = {5,6,7,8,9};
        while (i <= 5){
        int j = a[i%a.length];
        while (j > 1){
                System.out.print(j--);
                if (j == 2)
                break;
        }
         System.out.println("***");
        ++i;
        }
        double x = 7;
        double y = 8;
        double z = 9;
        System.out.println(x < y || y > z);
        System.out.println(x < y && y > z);
        System.out.println(x < y);
        System.out.println(x + y < z);
        System.out.println((x + y)-6 < z);
        }
}
```

## Task 9

Your professor expects only As, Bs, and Cs. In the following program, write a method called **getScores** that takes as input corresponding arrays **studentGrades** and **studentScores**. Write a method called **getScores** that assigns **index i** in studentGrades based on **index i** in **studentScores**. If a grade is **A**, assign **100**. If a grade is **B**, assign **90**. If a grade is **C**, assign **70**. If a grade is anything else, assign **0**.

| Sample Input | Sample Output |
|---|---|
| char[] studentGrades = new char[]{'A', 'A', 'A', 'B', 'C', 'U', 'Z'};<br>int[] studentScores = new int[7]; | Output expectation:<br>100<br>100<br>100<br>90<br>70<br>0<br>0 |

## Task 10

A. Write a method called **convertToCm()**, that takes as input a **type double** and **returns** the value converted from inches to centimeters.
**Hint**: There are 2.54 centimeters in an inch

| Sample Method Call | Output |
|---|---|
| double t = convertToCm(16);<br>System.out.println(t + " cm"); | 40.64 cm |

B. Create an **array** of **type double** of length **5** called **cheetos_inches**, that stores the length of each of the Cheetos **from the user**. Send the array of length in inches into a method called **findAvgCm()** that **returns** the average length of the Cheetos **in cm to 2 decimal places**. The method findAvgCm() uses **convertTocm()** to convert the length of each Cheetos **from inches to cm.**
**Note:** You must call the method written in [Method Task A], otherwise this task would be considered invalid.

| Sample Method Call | Output |
|---|---|
| Sample array:<br>double [] cheetos_inches = new double[]{10.0, 12.0, 14.0, 16.0, 18.0};<br><br>averageLength = findAvgCm(cheetos_inches);<br><br>System.out.println("The average Cheeto length is "+ averageLength +" cm"); | The average Cheeto length is 35.56 cm |

## Task 11

**A.** Write a method called **isVowel** which takes a string in its parameter and counts all the vowels in the String. If any vowel exists in the string then the method returns the **count**.

| Sample Input | Sample Output |
|---|---|
| The quick brown fox jumps over the lazy dog | Number of vowels in the string: 11 |

**B**. Write a method called **isConsonant** which takes a string in its parameter and counts all the consonants in the String. If any consonant exists in the string then the method returns the **count**.

| Sample Input | Sample Output |
|---|---|
| The quick brown fox jumps over the lazy dog | Number of consonants in the string: 24 |

**C.** Write a method called **vowel/consonantSum** which takes an array of strings in its parameter and returns the summation of the number of vowels/consonants.
**Note:** **You must call** the methods written in tasks **A/B**, otherwise this task will be **considered invalid**.

| Given Array | Sample Output |
|---|---|
| String [] names = {"Bob", "Alice", "Max", "Marry", "Rosy"};<br>System.out.println( "The total number of vowels in the array is:" + vowelSum(names));<br>System.out.println( "The total number of consonants in the array is:" + consonantSum(names)); | The total number of vowels in the array is: **7**<br><br>The total number of consonants in the array is: **13** |

## Task 12

```
public class P1{
    public static void main (String args[]){
        int a = 4, b = 7;
        System.out.println(methodA(a,b));
    }
    public static double methodA(int m, int n){
        int p = m+n-23, s = 0;
        if (p<0){
            System.out.println(p);
            System.out.println(methodB(p+10));
            s = methodB(p-10);
        }
        System.out.println(--s);
        return p*m+s;
    }
    public static int methodB(int r){
        int q = 6;
        System.out.println(++r + q);
        return q-r;
    }
}
```

# Task 13

```
public class P2{
    public static boolean met1(int n1, int n2){
        System.out.println("Method 1");
        int n = n1+n2;
        System.out.println(n);
        return met3(n, n2)>n1;
    }
    public static int met2(int n, String s){
        int p = 5;
        System.out.println("Method 2");
        System.out.println(met1(n,p));
        return s.length();
    }
    public static double met3(int n, int p){
        System.out.println("Method 3");
        System.out.println(n/p);
        return p;
    }
    public static void main (String args[]){
        System.out.println("Main Method");
        System.out.println(met2(6,"ABC"));
    }
}
```

# Task 14

```
public class P3{
    public static int calculate(int n) {
        if (n <= 0){
            return 4;
        }
        else if (n % 2 != 0){
            return n + calculate(n - 1);
        }
        else{
            return n * calculate(n - 2);
        }
    }
    public static void main(String[] args) {
        int result = calculate(8);
        System.out.println(result);
    }
}
```